

Oracle to SQL Migration

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1. Introduction

The document contains a step-by-step instruction for migration of a Therefore™ database from Oracle server to SQL server.

Scenario 1: Simple

Recommended for Therefore™ system with a smaller number of documents. This scenario requires exporting and importing all documents, so the procedure can take very long if there are a lot of documents in the system. Also, this scenario only allows migrating the latest version of each document, so the document history will be lost. Finally, this scenario does not allow running workflow instances to be migrated. Only workflow definitions can be migrated.

Scenario 2: Advanced

Recommended for systems whose Therefore™ tables are part of the SYSTEM schema on the Oracle server.

2. Scenario 1: Simple

Requirements

- Microsoft SQL Management Studio
- Therefore™ Solution Designer
- Therefore™ Document Loader
- Therefore™ Export Utility

Limitations

- Only the latest version of each Therefore™ document will be migrated.
- For workflows, only workflow definitions will be migrated, not instances.

2.1 Migration

On the Oracle Server

1. Export the Therefore™ configuration in the Therefore™ Solution Designer.
2. Export all Therefore™ documents using the Therefore™ Export Utility.

On the SQL Server

1. Perform the [additional steps below](#).
2. Import the Therefore™ configuration in the Therefore™ Solution Designer.
3. Import all Therefore™ documents using the Therefore™ Document Loader.

2.2 Additional Steps

1. Set the correct data types and primary keys for the Therefore™ tables on the SQL server (see the [comparison table](#)).
2. If the new Therefore™ Server computer name has changed, set correct computer name in the **TheServer** table.
3. If the new Therefore™ Server is in another domain, set the correct domain/computer name and group/user name in these tables: **TheServer**, **TheUser** and **TheGroup**.
4. Copy the Buffer and Cache folders from the old Therefore™ Server to the new one.
5. Install Therefore™.
6. Run the Therefore™ Configuration Wizard. Select the Therefore™ database on the SQL server.
7. In the Therefore™ Console, click on Tools and select **Optimize DB**.
8. Compare the number of rows in the migrated tables with the number of rows in the source tables and make sure all rows have been migrated.
9. Execute the following script on the SQL server:

```
WITH CTE AS (  
SELECT  
    st.Name AS TableName,  
    SUM(  
        CASE  
            WHEN (p.index_id < 2) AND (a.type = 1) THEN p.rows
```

```

ELSE 0
END
) AS RowCount
FROM sys.partitions p
INNER JOIN sys.allocation_units a ON p.partition_id = a.container_id
INNER JOIN sys.tables st ON st.object_id = p.Object_ID
INNER JOIN sys.schemas sch ON sch.schema_id = st.schema_id
GROUP BY st.name
)
SELECT * FROM CTE

```

10. Execute the following script on the Oracle server:

```

SELECT TABLE_NAME, TO_NUMBER (EXTRACTVALUE
(DBMS_XMLGEN.GETXMLTYPE('SELECT COUNT(*) c FROM ' ||
TABLE_NAME), '/ROWSET/ROW/C')) RowCount FROM USER_TABLES ORDER BY TABLE_NAME;

```



It's not a problem if the Therefore™ database on the SQL server contains more rows than on the Oracle Server.

3. Scenario 2: Advanced

Requirements

- Microsoft SQL Management Studio
- Oracle Client software. This must be the same bit version (32 or 64 bit) as the SQL Management Studio.
- Therefore™'s TheSchema tool. The tool must have the same version as the Therefore™ Server on the SQL Server. Contact Therefore™ support to get this tool.

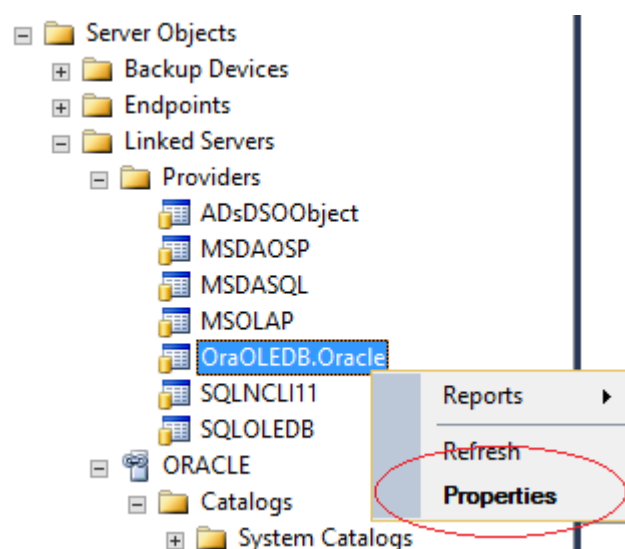
3.1 Preparation

1. Install the Oracle client on the machine that should be used for migration. This does not necessarily have to be the Therefore™ Server.
2. Configure the Oracle client to connect to the Oracle server and make sure the connection to the Oracle server can be established.
3. Create a Linked Server for Oracle on the SQL server using the statement below (please read the script comments):

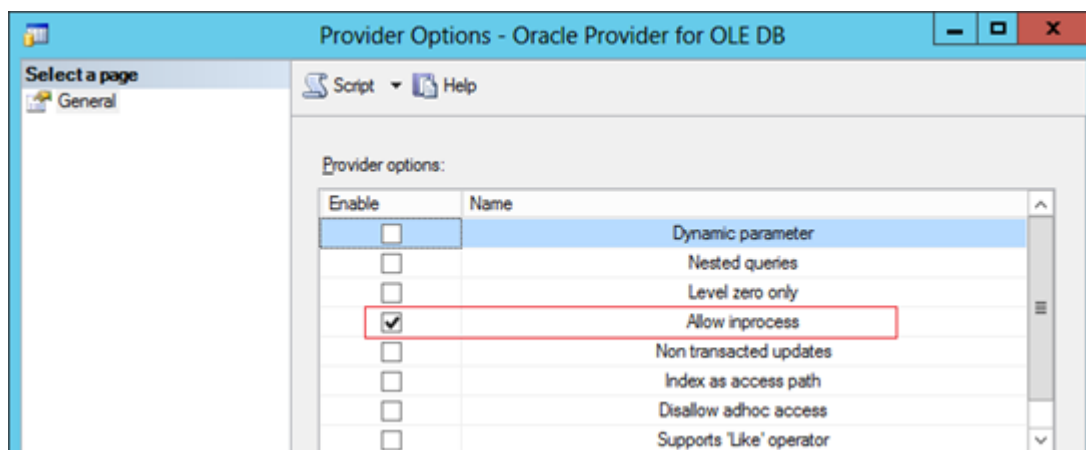
```
USE [master]
GO
```

```
/* Replace <oracle_service> with a service name of Oracle client*/
EXEC master.dbo.sp_addlinkedserver @server = N'ORACLE', @srvproduct=N'Therefore',
@provider=N'OraOLEDB.Oracle', @datasrc=N'<oracle_service>'
/* Replace <oracle_user> with a username of the Oracle user*/
/* Replace <password> with a real password*/
EXEC master.dbo.sp_addlinkedsrvlogin
@rmtsrvname=N'ORACLE',@useself=N'False',@locallogin=NULL,@rmtuser=N'<oracle_user>'
,@rmtpassword='<password>'
```

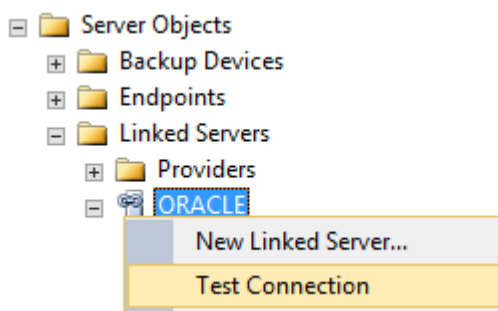
4. Expand the Providers item under **Linked Servers**, right-click on **OraOleDB.Oracle** and select Properties.



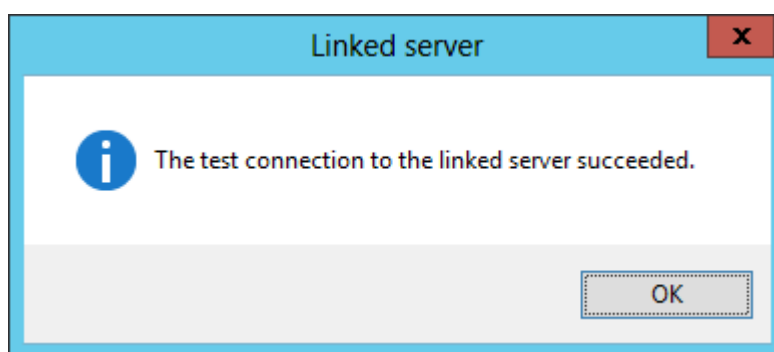
5. Enable the option **Allow inprocess** and save your changes.



6. Right-click on the linked server "ORACLE" and choose **Test Connection**.



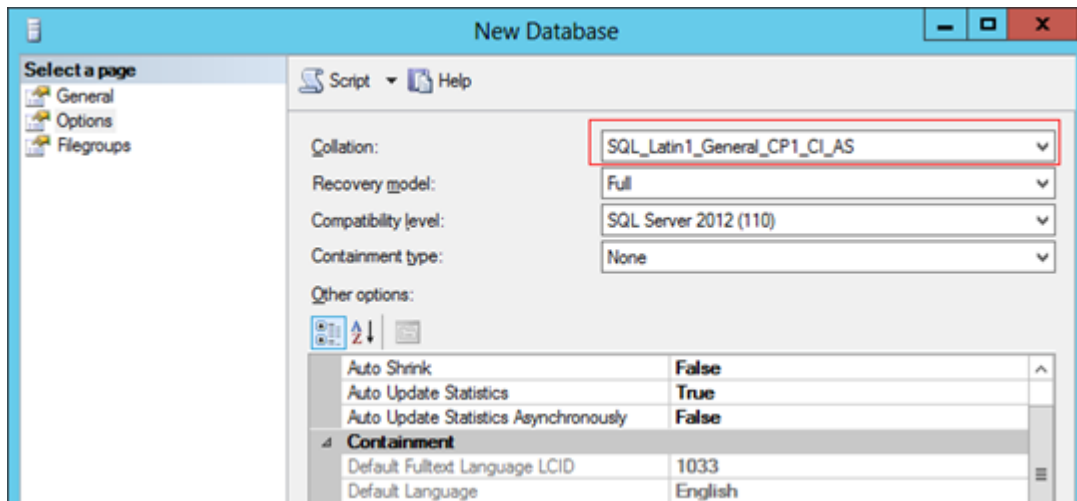
7. If the linked server was configured correctly, the dialog below will appear.



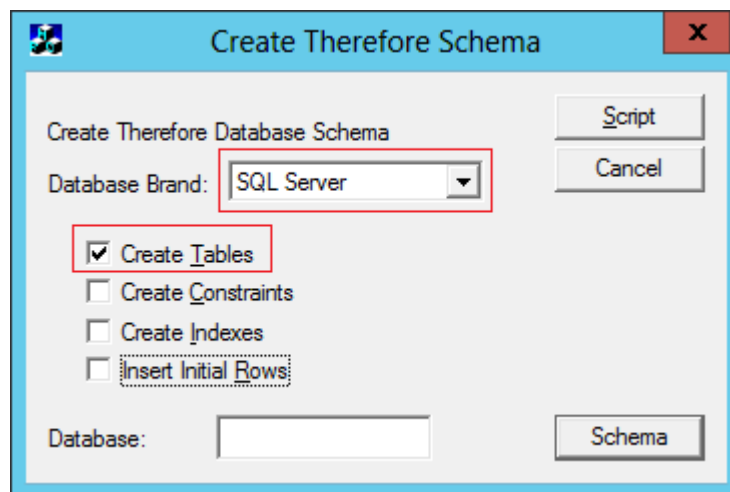
8. Install the Therefore™ Server as usual, using the same Therefore™ version as on the Oracle server. Do NOT run the Therefore™ Configuration Wizard.

3.2 Migration

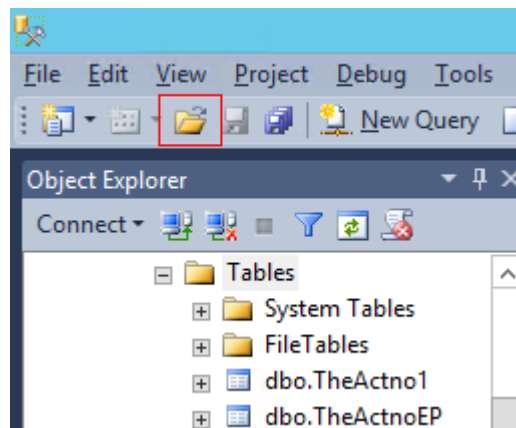
1. Create an empty Therefore™ database on the SQL server.



2. If no special database collation is required, make sure the database collation is set to **SQL_Latin1_General_CP1_CI_AS**.
3. Copy the TheSchema tool to the Therefore™ folder on the Therefore™ Server. If the tool version is 32-bit, copy it to C:\Program Files (x86)\Therefore. If 64-bit, copy it to C:\Program Files\Therefore.



4. Launch the TheSchema tool. Make sure the user who launches the tool has Write access to the folder the TheSchema tool is located. Note that the tool can only save scripts to the Therefore folder.
5. Choose **SQL Server** as the Database Brand and only check the box **Create Tables**.
6. Click the **Script** button and save the script.
7. Execute the script created by the TheSchema tool on the Therefore™ database.
8. In SQL Management Studio, click on the **Open File** icon.



9. Open and execute the following scripts one after another in the following order. Click on the tag next to each title to open the script file. You may download these scripts from our Extranet page: [Sales/Technical Info – Therefore](#)
 1. FillDefaultTables.sql
 2. FillTablesCLOBdatatype.sql
 3. CreateNonDefaultTables.sql
 4. FillTheRecycleBin.sql
 5. CreateTheKeywords.sql
 6. CreateTheKeywordTemplateType.sql
 7. CreateTheCat.sql
 8. CreateTheCase.sql
 9. CreateTheIXtable.sql

The scripts will create a temporary table, MigrateLog, in the Therefore™ database on the SQL server. This table contains information about the execution of each statement. Once the migration is finished, the table must be manually deleted from the Therefore™ database on the SQL server.

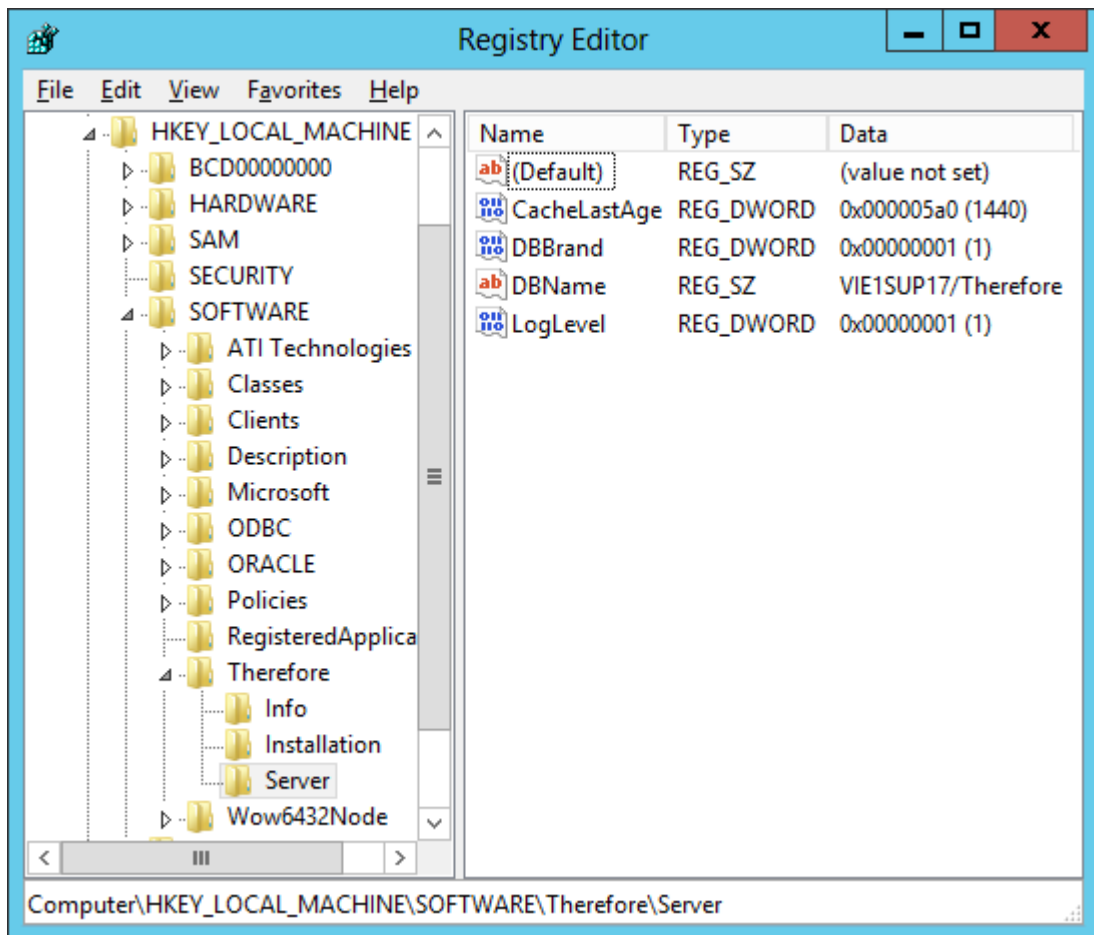
10. To migrate custom tables from Oracle to SQL, execute the following statement on the SQL server:

```
SELECT * INTO <TABLE_NAME> FROM OPENQUERY(oracle, 'SELECT * FROM <TABLE_NAME>')
```

Replace **TABLE_NAME** with the real name of a table on the Oracle server.

11. If the Therefore™ Server will be installed on a new machine, you need to modify the **TheServer** table. Set the value for the column "Computer" to the new computer name of the Therefore™ Server.
12. Create the following registry keys on the Therefore™ Server under **HKLM\SOFTWARE\Therefore\Server**

```
DBBrand (DWORD) = 1
DBName (String) = "<server>/<db_name>"
```

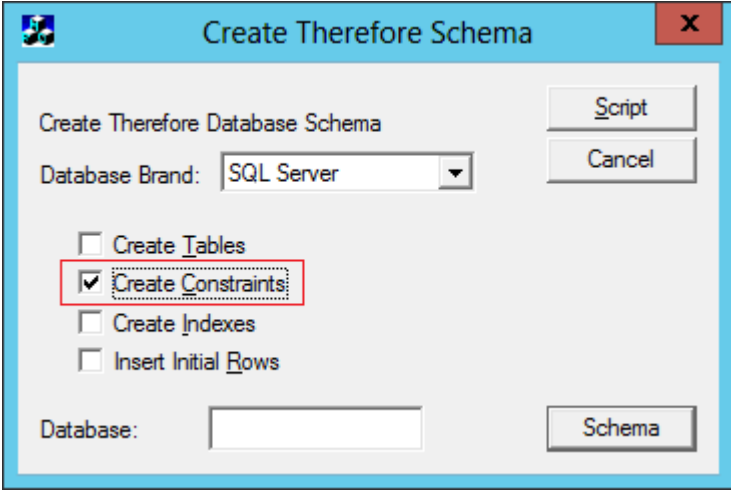


13. If you need to use SQL authentication, do the following:
- Open the Therefore™ Solution Designer on the Therefore™ Server connected to Oracle.
 - Right-click the Therefore™ root node, choose **Settings** and switch to the Database tab.
 - Input the username and password for the SQL user and click OK to save the changes. Keep the Solution Designer open.
 - Export the registry key **HKLM\SOFTWARE\Therefore\Server**.

- e. In the Solution Designer on the Therefore™ Server connected to Oracle, right-click the Therefore™ root node, choose **Settings** and switch to the Database tab.
- f. Clear the username and password fields and click OK to save changes.
- g. Import the registry settings on the Therefore™ server connected to the SQL server.

14. Launch the TheSchema tool again.

15. Choose **SQL Server** as the Database Brand and only check the box for **Create Constraints**.

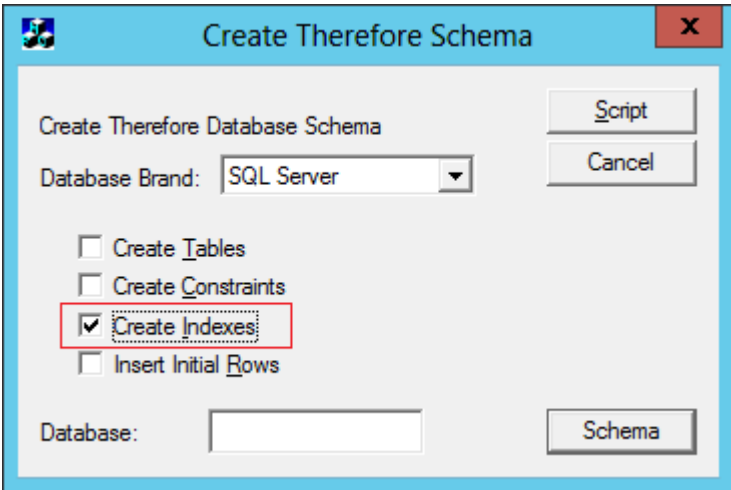


The screenshot shows the 'Create Therefore Schema' dialog box. The title bar is blue with a close button (X). The main area is light gray. At the top, it says 'Create Therefore Database Schema'. Below that is a dropdown menu for 'Database Brand' set to 'SQL Server'. To the right are 'Script' and 'Cancel' buttons. In the center, there are four checkboxes: 'Create Tables' (unchecked), 'Create Constraints' (checked and highlighted with a red box), 'Create Indexes' (unchecked), and 'Insert Initial Rows' (unchecked). At the bottom, there is a text field for 'Database:' and a 'Schema' button.

16. Click the **Script** button and save the script.

17. Execute the script created by the TheSchema tool on the Therefore™ database and check the output window below the query dialog. If the process was successful, you will not see any errors.

18. In the TheSchema tool, only check the box for **Create Indexes**.



The screenshot shows the 'Create Therefore Schema' dialog box. The title bar is blue with a close button (X). The main area is light gray. At the top, it says 'Create Therefore Database Schema'. Below that is a dropdown menu for 'Database Brand' set to 'SQL Server'. To the right are 'Script' and 'Cancel' buttons. In the center, there are four checkboxes: 'Create Tables' (unchecked), 'Create Constraints' (unchecked), 'Create Indexes' (checked and highlighted with a red box), and 'Insert Initial Rows' (unchecked). At the bottom, there is a text field for 'Database:' and a 'Schema' button.

19. Click the **Script** button and save the script. Execute the script created by the TheSchema tool on the Therefore™ database and check the output window below the query dialog. If the process was successful, you will not see any errors.
20. Start the Therefore™ Server Service. If you see any errors related to the Therefore™ Server Service start up, analyze and fix the problem and restart the Therefore™ Server Service. Repeat the steps until you do not see any errors. If errors like this: "Failed to load information from database: Category - ID=..." are found in the Windows application event log, you can ignore these errors for the time being.
21. In the Therefore™ Console, click on Tools and select **Optimize DB**.
22. Compare the number of rows in the migrated tables with the number of rows in source tables and make sure all rows have been migrated. Execute the following script on the SQL server:

```
WITH CTE AS (
SELECT
    st.Name AS TableName,
    SUM(
        CASE
            WHEN (p.index_id < 2) AND (a.type = 1) THEN p.rows
            ELSE 0
        END
    ) AS RowCount
FROM sys.partitions p
INNER JOIN sys.allocation_units a ON p.partition_id = a.container_id
INNER JOIN sys.tables st ON st.object_id = p.object_id
INNER JOIN sys.schemas sch ON sch.schema_id = st.schema_id
GROUP BY st.name
)
SELECT * FROM CTE
```

23. Execute the following script on the Oracle server:

```
SELECT TABLE_NAME, TO_NUMBER (EXTRACTVALUE
(DBMS_XMLGEN.GETXMLTYPE('SELECT COUNT(*) c FROM ' ||
TABLE_NAME), '/ROWSET/ROW/C')) RowCount FROM USER_TABLES ORDER BY TABLE_NAME;
```



It's not a problem if the Therefore™ database on the SQL server contains more rows than on the Oracle Server.

24. In the Therefore™ Console, click on Verify and select **Verify Documents**.
25. If necessary, export the content of the **MigrateLog** table and then delete it.

4. SQL-Oracle Column Datatype Comparison

ORACLE				SQL Server		
Data Type	Data Length	Data Precision	Data Scale		Data Type	Data Length
DATE	any	NULL	NULL	➔	Datetime	
TIMESTAMP	any	NULL	3	➔	Datetime	
NUMBER	22	15	2	➔	Decimal	15,2
NUMBER	22	NULL	NULL	➔	Int	
NUMBER	22	1	0	➔	Tinyint	
NUMBER	22	5	0	➔	Smallint	
NVARCHAR2	any	NULL	NULL	➔	Nvarchar	any
RAW	any	NULL	NULL	➔	Uniqueidentifier	